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Paper Code: TMA-101

End Semester Examination, 2019
B.Tech – Ist year Ist Semester
Engineering Mathematics- I

Time: Three Hours

MM: 100

Note:

1. This question paper contains five questions.
2. All questions are compulsory.
3. Instructions on how to attempt a question are mentioned against it.
4. Total marks assigned to each question are twenty.

Q1 (Attempt any two questions of choice from a, b and c) (2x10=20 Marks).

(a) If $y = \frac{1}{1+x^2}$, Find y_n

(b) If $y = \log(x + \sqrt{1+x^2})$, Prove that $(x^2+1)y_2 + x y_n = 0$

(c) If $y = e^{3x} \sin(3x+3)$, Find y_n

Q2 (Attempt any two questions of choice from a, b and c) (2x10=20 Marks).

(a) If m, n, p are the roots of the equation $(\lambda-x)^3 + (\lambda-y)^3 + (\lambda-z)^3 = 0$ in λ find $\frac{\partial(m,n,p)}{\partial(x,y,z)}$

(b) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$

(c) A rectangular box, which is open at the top has a capacity of 256 cubic feet determine the dimensions of the box such that the least material is required for the construction of the box. Use Lagrange's method of multipliers to obtain the solution.

Q3 (Attempt any two questions of choice from a, b and c) (2x10=20 Mark s).

(a) Prove that $A = \begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ is a Hermitian matrix.

(b) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 0 \\ -2 & 1 & 4 \end{bmatrix}$.

(c) Find eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$.

Q4 (Attempt any two questions of choice from a, b and c) (2x10=20 Marks).

(a) Evaluate $\iint_R xy dx dy$ over the region in the positive quadrant for which $x + y \leq 1$.

(b) Evaluate $\iint_R xy dx dy$, where R is the quadrant of the circle $x^2 + y^2 = a^2$ where $x \geq 0$ and $y \geq 0$.

(c) If $\phi = 3x^2y - y^3z^2$, find the grad ϕ at the point (1, -2, -1).

Q5 (Attempt any two questions of choice from a, b and c) (2x10=20 Marks).

(a) Using Green's theorem, evaluate $\int_C (x^2y dx + x^2 dy)$ where c is boundary described counterclockwise of the triangle with vertices (0, 0), (1, 0) and (1, 1).

(b) Find the total work done by a force $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$, moving a point at (0,0) to (a,b) along the rectangle bounded by the lines $x=0$, $y=0$, $x=a$ and $y=b$.

(c) If $\vec{v} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}}$, find the div $\vec{v}$